

Arctic Sea Ice Loss Case Study Rubric

DS 4002 – Spring 2026 – Kendall Freese

Due: TBD

Submission format: Upload a link to your repository on Canvas and upload your written document

Individual Assignment

General Description: Submit to Canvas a link to your case study repository and upload your written document.

Why am I doing this? This case study serves as an opportunity to convey all that you have learned in DS 4002 and at UVA. Through this case study, you will be able to exemplify your technical, qualitative, and time management skills that you have gained thus far at the university. This is your time to highlight your strengths and identify your weaknesses.

What am I going to do? First you will review the given github repository found at: . <https://github.com/KendallFreese/CS3-DS4002/tree/main>. You will read all supporting materials and documents to familiarize yourself with the material. You will then download the given dataset. Consult the given code in the “data” folder to aid in your own analysis, be mindful that this analysis was from the past, and put your own spin on certain technical components. Produce results and interpret them meaningfully.

Tips for success:

- Review all materials on the topic and technical goals to familiarize yourself with the future work
- Question initial results and be critical of outputs
- Interpret results along the way to make sense of the data and create meaningful conclusions
- Utilize resources when stuck and need clarification

How will I know I have Succeeded? You will meet expectations on the case study by following the rubric below:

Formatting	• Written Document ○ Submit written document as a .pdf file to Canvas • Code ○ Submit code and all data analysis to a public GitHub repository ○ The GitHub should be titled “CS_Study_[FirstName]_[LastName]” • References ○ All references should be included in a separate page at the bottom of the Written Document ○ IEEE citation style
Written Document	<u>Goal:</u> Create a written document that summarizes your model and results and provides meaningful interpretation and next steps.

(.pdf)	● Method: ○ Introduce the goal of the project ○ Outline analysis steps and choices ○ Justify decisions made to model design and analysis plan ● Results: ○ Analyze visualizations from EDA ○ Provide results and conclusions for historical time series analysis ○ Provide visualizations of the three models and report performance metrics ○ Explain and interpret the results of the models ● Interpretation: ○ Explain your results and choice of best model ○ Interpret the near-zero value and what it means for the future ● Reflection ○ Reflect on what you succeeded in for this project ○ Reflect on what you could have done better ○ What changes could improve your model?
Code	<u>Goal:</u> ● Consult the DS4002Project2Code.Rmd in the “data” folder as a guide for analysis ● Exploratory Data Analysis ○ Create plots for: monthly average sea ice extent, annual average sea ice extent, September ice extent, March ice extent, average extent by month ● Linear trend analysis ○ Plot annual extent with linear trend ○ Visualize seasonal difference analysis (March vs. September extent), ● Quadratic trend analysis: ○ Test for acceleration ○ Visualize the quadratic trend of annual extent ● Volatility analysis ○ Visualize extent volatility by decade ○ Plot 10-year rolling standard deviation of annual extent ● Forecasting ○ Chronological train/test split ○ Build ARIMA model ○ Fit forecasting models: linear, quadratic, and ARIMA ○ Determine best model by comparing performance metrics ● Near-Zero ○ Use best performing model to calculate near-zero ● Comment your code ● All scripts should be uploaded to its designated folder
References	● All additional references (beyond given reference materials) should be included in a separate page at the bottom of the Written Document ● IEEE citation style

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from [Streifer & Palmer \(2020\)](#).